SOLVING LINEAR EQUATIONS

Goal: The goal of solving a linear equation is to find the value of the variable that will make the statement (equation) true.

Method: Perform operations to both sides of the equation in order to isolate the variable.

Addition and Subtraction Properties of Equality:

Let , , and represent algebraic expressions. 1. Addition property of equality:

If ,

then

2. Subtraction property of equality:

If ,

then

Multiplication and Division Properties of Equality:

Let , , and represent algebraic expressions. 1. Multiplication property of equality: If ,

then

2. Division property of equality:

If ,

then (provided 0)

Clearing Fractions or Decimals in an Equation:

When solving an equation with fractions or decimals, there is an option of clearing the fractions or decimals in order to create a simpler equation involving whole numbers.

1. To clear fractions, multiply both sides of the equation (distributing to all terms) by the LCD of all the fractions.
2. To clear decimals, multiply both sides of the equation (distributing to all terms) by the lowest power of 10 that will make all decimals whole numbers.

Steps for Solving a Linear Equation in One Variable:

1. Simplify both sides of the equation.
2. Use the addition or subtraction properties of equality to collect the variable terms on one side of the equation and the constant terms on the other.
3. Use the multiplication or division properties of equality to make the coefficient of the variable term equal to 1.
4. Check your answer by substituting your solution into the original equation. Note: If when solving an equation, the variables are eliminated to reveal a true statement such as, 13 13, then the solution is all real numbers. This type of equation is called an identity. On the other hand, if the variables are eliminated to reveal a false statement such as, 7 3, then there is no solution. This type of equation is called a contradiction. All other linear equations which have only one solution are called conditional.

Examples:

A. 5 2 Check:

5 5 7

B. 3 7 3 3

10

7 5 2

2 2 (Solution Checks) Check:

10 3 7

7 7 (Solution Checks)

C. 3 24

24

3 3

3

8

D. 7 28

7 28

7

7

4

E. 4

· · (Multiply by the reciprocal)

6

F. ! 2

5

"

1· 5 2 5

" 10

Identity Example:

G. 2 6 3 2

2 6 3 6

2 6 2 6

2 2

6 6 True Statement

Solution: all real numbers

I. 10 5 3 4 2 7 10 5

3 12 2 14 10 5 26

10 6 26

10 10

6 36

#$

#

# #

6

J. $

! $

2 (LCD is 10)

10 2

5 4

2 2 10

1· 2

Check:

3 8 24

24 24 (Solution Checks)

Check:

7 4 28

28 28 (Solution Checks)

Check:

2

6

3 1 4

4 4 (Solution Checks)

Check:

10

5 2

2 2 (Solution Checks)

Contradiction Example:

H. 5 3 4 2

5 3 4 8

5 3 5 8

5 5

3 8 False Statement No Solution

Check:

10 5 6 3 6 4 2 6 7 10

30 3 2 2 13

20 6 26

20 20 (Solution Checks)

K. 0.05 0.25 0.2

100 0.05 0.25 0.2 100

5 25 20

_25 _25
10 10
5 1· 4 2 20
2 2 5 4 20 2 4 5 20 20 3
16 20
_16 _16
$

!$!
! !
1
Check:
0.05 1 0.25 0.2
0.05 0.25 0.2
0.2 0.2 (Solution Checks)

1. 4 4
2. 6 7
3. 4 7
4. $ 9
5. 2 4 8
6. 14 3 2
7. 8 3 19
8. 6 9
9. 12
10. 3 2 6
11. 3 2 8 12
12. 4 6 2 0
13. 3 2 6 15
14. 4 2 3
15. 27 46 2
16. 4 6 7 9 18 17. 4
3 2 10
18. 3 3 2 1 19. 9
6 3 30
20. 2 2 3 6 21. 2
6 3 9 3 22. 5 3 2
10
23. 3 12 24 9 24. 2 4
3 5 25. 4 2 3 4 8 8 26.
6 11 6 5 27. 2 7 6 9
4 28. 5 3 4 6 20 9 29. 4
3 5 0 30. 2 4 6 2
3 31. , $
32. $ (
#
33.) ( )!
34. *+
35. !
36. ' (9 '!

LINEAR EQUATIONS PRACTICE

37. _ '!
38. 2
39. $ $ #
40. ! 26
41. (* !
42. #)
43. $+
, '
44. $+!
'
#
45. 6 $ $_
46. $ $
47. ! 3
48. !
49. $
$!#
50. $ 12
51. 12
52. 2 16
53. 2 14
54. ' 8
55. ' 2
56. $ 4
57. 26
58. 3 15
59. 4 32
60. 5
61.) 5
62. ! 15
63. # ! 30
64. # ! 90
65. 4
66. ' # 168
67. # 2
68.) ! 45
69.) 36
70. 4 4 40
71. 9 7 34
72. '_" 6 8
73. 10 6
74. 2 4 1
75. 9 9 9
76. $ $! 2
77. $, $) 2
78. $ $) 6
79. $ _ $) 1
80. !)" 4 6
81. 0.4 3.5
82. 5 3 45
83. 3 7 9
84. 4 6 12
85. 8 2 5 6
86. 2 7 4 9
87. 1 3 2 3 2 88. 3 4 2
5 3 89. 3.65 7.4 1.12 21.76
90. 8 3 2 6 3 4 91. 10
3 10 13 3 7 92. 6 3 5
1 3 2 93. 10 5 3 4 2
7 94. 9.2" 4.3 50.9

95. 0.05/ 0.2 0.15/ 10.5 96. 0.25 60 0.10 0.15 60

97. 0.5 30 87 1.50 43.5 98. 0.4 " 10 0.6" 2 99. 21.11

4.6 10.91 35.2 100. 0.125 0.025 5 1

LINEAR EQUATIONS PRACTICE ANSWERS

1. 1
2. 13
3. 11
4. 27
5. 2
6.
7. 2
8. 3
9. 12
10. 4
11. 6
12. 3
13. !
14. 5
15. 19
16. 1
17. 0
18. !
19. 3
20. (,
21. 0
22. 1
23. No Solution 24. !
25. All real numbers
26.
27. No Solution 28. All real numbers 29. '
30. (,
31. 12
32. 5
33. 2
34. / 0
35. 5
36. 2 30
37. 1 5
38.
39. 3
40. 30
41. / 30
42. 1
43. 1
44. 4
45. 16
46. !
47. !-
48. " 1
49. 14
50. 48
51. 12 52. 8 53. 7 54. 56 55. 14 56. 8 57. 26 58. 5 59. 8 60. 15 61. 45 62. 9 63. 25 64. 75 65. 12 66. 144 67. 12 68. 25 69. 81 70. 9 71. 3 72. " 16 73. 4 74. # 75.
76. 15
77. 63
78. 27
79. 72
80. " 18
81. 2.5
82. 12
83. 10
84. 3
85.)
86. -
87. 4
88. 6
89. 3 2.3
90. All real numbers
91. No Solution
92. 5
93. 6
94. " 6
95. / 107
96. 120
97. All real numbers
98. " 2
99. 1 3
100. No Solution